

Project- Final Report

Predictive Analytics– BIS404

**Sales Forecasting for Business Planning:
Weekly Sales Forecasting for Walmart Using
ARIMA and LSTM Models**

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Abstract:

This project applies predictive analytics techniques to forecast Walmart's future weekly sales using time-series modeling. Accurate sales forecasting is critical for retail operations, enabling organizations to optimize inventory levels, avoid stockouts, reduce excess storage costs, and plan promotions more effectively. In this study, two forecasting approaches were implemented and compared: a traditional ARIMA model and a deep learning-based Long Short-Term Memory (LSTM) network. The Walmart Weekly Sales dataset contains more than 420,000 records; however, a representative 1000 row subset was used for implementation due to computational constraints and to demonstrate the modeling workflow clearly.

The methodology included data cleaning, frequency standardization, descriptive visualization, and an 80/20 train-test split. The ARIMA model was trained on the time-indexed series, while the LSTM model was constructed using scaled sliding windows and trained for 30 epochs. Model performance was evaluated using MAE, MSE, and RMSE. Results show that both models captured the overall sales trend, with the LSTM model showing slightly smoother predictions, while ARIMA performed competitively despite the limited dataset size.

The findings highlight the value of time-series forecasting for business decision-making and demonstrate the trade-offs between statistical and deep learning approaches. This project provides practical insights into predictive modeling and supports the design of data-driven retail strategies.

Introduction:

Forecasting has become an essential capability to maintain operational efficiency and ensure customer satisfaction in today's data-driven retail environment. Predictive analytics plays a key role in large retailers such as Walmart to anticipate fluctuations in demand, streamline their inventory planning, and optimize supply chain processes. Weekly sales data would act as the key indicator of consumer behavior, seasonal patterns, and effects from external factors like economic conditions or holidays. For retail businesses operating on the scale that Walmart does, even a relatively small improvement in forecasting accuracy could save huge amounts of money. Therefore, forecasting future weekly sales has a direct influence on business performance, resource allocation, staffing, and long-term planning.

Time-series forecasting provides a sound framework for analyzing historical patterns and making reliable predictions. The traditional statistical models, such as ARIMA, have enjoyed widespread usage, owing both to their interpretability and their performance in capturing linear dependencies of temporal data. However, in modern business environments, complex nonlinear patterns are more common and obviously require advanced modeling techniques. Deep learning models, particularly Long Short-Term Memory (LSTM) networks, have emerged as powerful tools for modeling nonlinear trends and long-range dependencies in sequential datasets. And By comparing ARIMA and LSTM models, this project aims to highlight the strengths and limitations of each approach within the context of real-world retail forecasting.

The data set used in this project comes from the publicly available Walmart Sales Forecasting Dataset, containing over 420,000 records of historical weekly sales for various stores and departments. Even though the full dataset contains voluminous contextual features, including holidays, fuel prices, temperature, CPI, and unemployment, this project will use a representative subset of 1000 weekly sales records for implementation. This is appropriate for computational constraints and also suitable for an education-focused objective in core predictive analytics. This subset retains meaningful variability that enables one to clearly compare models and explain methodologies.

The motivation for this work is inspired by the growing use of AI and machine learning as drivers of changing retail operations. Retailers continuously seek data-driven solutions to optimize supply chain decisions, reduce operational costs, improve product availability, and enhance strategic planning. This research focuses on evaluating two forecasting models and illustrates ways predictive analytics can support business decision-making, offering practical insights about choosing models based on characteristics of the data and organizational needs.

This project is guided by the following research questions:

- (1) Can ARIMA and LSTM models accurately predict future weekly sales using historical data?
- (2) Which of the two models yields more accurate and stable forecasts?
- (3) How can improved forecasting accuracy contribute to better business planning and operational efficiency?

The main objective of this study is to build, evaluate, and compare ARIMA and LSTM forecasting models using cleaned and preprocessed Walmart weekly sales data. Additional objectives include data visualization, train-test splitting, model training, and performance evaluation using MAE, MSE, and RMSE metrics. The key value delivered by the project is in showing real-life application of different forecasting methods and providing guidelines on how predictive analytics may contribute to better decision-making in a large retail context.

Related Work:

Sales forecasting has been one of the most enduring topics of interest in operations management, economics, and data science because it plays a critical role in supporting strategic and tactical decision-making. For time-series forecasting, traditional statistical methods like ARIMA remain widely popular owing to their interpretability, conceptual mathematical underpinning, and great performance in stationary datasets. According to Shmueli et al. (2025), ARIMA models remain foundational tools in business analytics, offering effective mechanisms for catching linear trends and short-term dependencies. Many studies demonstrate the suitability of ARIMA in retail forecasting tasks whose historical sales patterns show stable seasonal or cyclical behavior.

However, as retail environments grow increasingly complex and influenced by nonlinear interactions, deep learning models, especially Long Short-Term Memory (LSTM) networks, have been of considerable interest. LSTM models have the advantage of learning long-range

temporal dependencies due to their gated architecture, which overcomes vanishing gradient limitations that affect traditional recurrent neural networks. Research by Shmueli et al. (2020) underlines that deep learning methods outperform classical statistical models when the underlying time-series data includes nonlinear fluctuations, irregular seasonality, or complex interactions with external variables.

Recent literature on retail forecasting tends to compare ARIMA and LSTM models in order to evaluate the performance advantages of each model type. Many studies show that ARIMA does well on smaller or less volatile datasets, while large datasets with rich dynamics are where LSTM models shine. For example, work using the Walmart Sales Forecasting dataset from Kaggle has illustrated that LSTM models will generally produce smoother, more accurate long-horizon predictions when holiday indicators, temperature, and economic variables are included. In contrast, ARIMA modeling is chosen in favor of fast baselines, lower computational cost, and higher interpretability, yielding added value for managerial decision-making. Table 1 summarizes key distinctions between ARIMA and LSTM models, illustrating their relative advantages based on prior research.

Criterion	ARIMA	LSTM
Pattern type best suited	Linear trends and seasonal data	Nonlinear and complex temporal patterns
Data size requirement	Performs well with small datasets	Require larger datasets to generalize
Interpretability	High (clear parameters: p, d, q)	Low (black-box deep learning model)
Computational cost	Low	High
Ability to capture long-term dependencies	Limited	Strong (memory cells & gates)
Typical business use cases	Baseline forecasting, short-term predictions	Advanced forecasting, volatile environments

Table 1. Comparison of ARIMA and LSTM Models Based on Prior Literature

Overall, existing studies validate the appropriateness of the evaluation of both statistical and deep learning methods for retail sales forecasting. By applying ARIMA and LSTM to the Walmart dataset, this work extends established methodologies while illustrating the practical differences between classical and modern predictive models in business analytics.

Methodology:

This project follows a structured methodology designed to transform raw Walmart sales records into a time-series format suitable for forecasting and to evaluate two predictive modeling approaches.

For data preparation: The original Walmart dataset contains multiple sales entries for each date for various stores and departments. To develop a single forecasting target, all records that

shared the same date were reduced to one value representing the total weekly sales. The resulting data was chronologically ordered and aligned to a regular frequency to ensure that the time-series structure was complete and unbroken. From this, 1,000 weekly observations were taken to strike a good balance between computational feasibility and data richness, which would enable pattern identification.

The **exploratory analysis:** Exploratory visualizations include line plots, histograms, scatterplots, and boxplots that show the distribution and temporal behavior of the weekly sales. These visual insights supported the identification of trends, fluctuations, and variability that informed model expectations.

model selection: Two forecasting frameworks were selected, in view of their conceptual strengths:

ARIMA: A traditional statistical model that is suitable for representing linear patterns and short-run dependencies in time series data.

LSTM: A deep learning model able to learn nonlinear relationships and long-range temporal dynamics; thus, suitable for more complicated retail sales behavior.

Last thing is **comparative evaluation:** Both models were evaluated on their ability to generate accurate forecasts and capture sales dynamics. The comparison assesses differences between linear and nonlinear modeling approaches and supports the project's objective of identifying the more effective forecasting technique for aggregated retail sales.

Dataset:

It is based on the Walmart Weekly Sales Forecasting Dataset published on *Kaggle by Ahmedov (2023)*. The dataset contains extensive historical sales records collected from multiple Walmart stores and departments between 2010 and 2012. In its full form, this dataset includes over 420,000 rows of data and multiple features related to store information, department identifiers, weekly sales, holiday indications, economic variables such as CPI and unemployment, temperature, and fuel prices. This richness of the dataset makes it one of the most widely used benchmark datasets in time series forecasting studies and predictive analytics education.

Although the full data set offers rich contextual information, this model concerns the relationship between date and weekly sales; thus, only a representative subset of 1000 weekly observations is chosen for implementation. This subset allows for computational efficiency and easier demonstration of the modeling pipeline while still being representative for meaningful variation in sales behavior. Using a smaller sample also aligns classroom execution constraints and allows training ARIMA and LSTM models without high-performance hardware.

forecasting models like LSTM require high computational power when trained on large datasets. For the purpose of this academic project, the Justification for Using a 1000-row subset is the Need for clear demonstration of ARIMA and LSTM workflows and Ability to visually inspect results more easily. Despite the reduced size, the subset preserves the essential temporal behavior required for evaluating forecasting models.

Experimental Setup:

All experiments were conducted in Python using widely adopted data science libraries. The tools used include **Pandas** for data preparation, **NumPy** for numerical operations, **Matplotlib** for visualization, **Statsmodels** for ARIMA modeling, **scikit-learn** for scaling and evaluation, and **TensorFlow/Keras** for constructing and training the LSTM model. This environment ensures full reproducibility of the forecasting process.

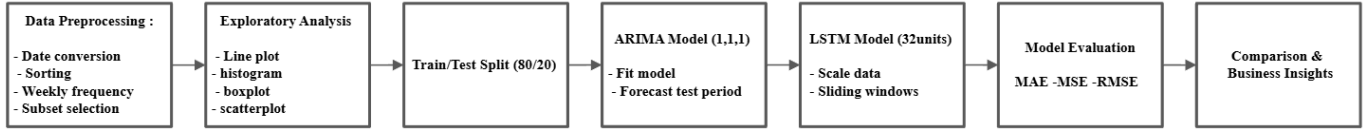


Figure 1. Experimental Setup Workflow Diagram

And since the Walmart dataset contains multiple entries per date (representing different stores and departments), the following preprocessing was performed **before modeling**:

1. **Date Conversion:** The Date column was converted into Python datetime objects to ensure chronological accuracy.
2. **Sorting:** The dataset was sorted by date to maintain correct temporal order.
3. **Aggregation by Date :** Because the dataset includes multiple records per date, all sales values belonging to the same date were **summed up** to produce a single weekly sales figure. This step ensures that each index corresponds to exactly one value.
4. **Frequency Standardization:** Weekly frequency was enforced using `.asfreq('W-FRI')`, which aligns the index to consistent weekly intervals. Any missing values created by this step were filled using forward-fill and back-fill.
5. **Handling Missing Weeks:** Any missing weekly entries created during resampling were filled using forward-fill and back-fill techniques.
6. **Subset Selection:** the first **1,000 aggregated weekly records** were used to allow more stable model training and analysis.
7. **Indexing:** The aggregated date column was set as the time-series index.

This ensured that the resulting time series had **unique dates**, consistent weekly intervals, and a continuous temporal structure.

The aggregated time series was split chronologically using an **80/20 rule**:

Subset	Number of Rows	Percentage
Training Set	114	80%
Test Set	29	20%

Table 2. Train–Test Split

This approach ensures that the models train on past observations and are evaluated on **future unseen data**, preventing information leakage.

Before modeling, several visualizations were generated to understand sales behavior:

Line Plot: Shows the overall pattern of total weekly sales across the first 1,000 records.

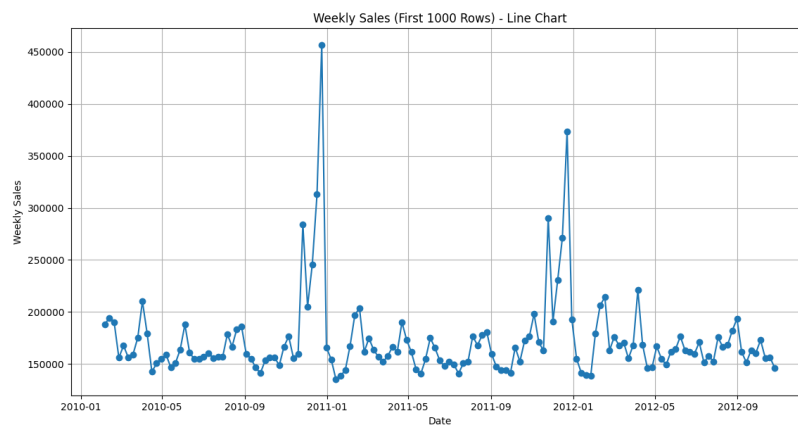


Figure 2 — Weekly Sales Line Plot

Histogram: Displays the distribution of aggregated weekly sales values.

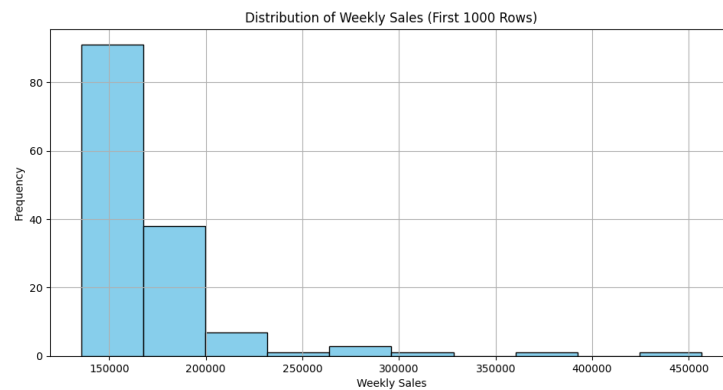


Figure 3 — Histogram

Boxplot: Highlights the presence of variability or outliers in sales.

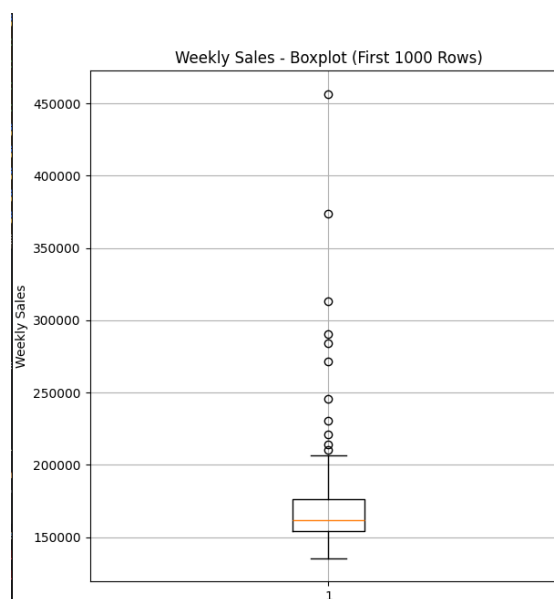


Figure 4 — Boxplot

Scatter Plot: Shows point-level fluctuations over time.

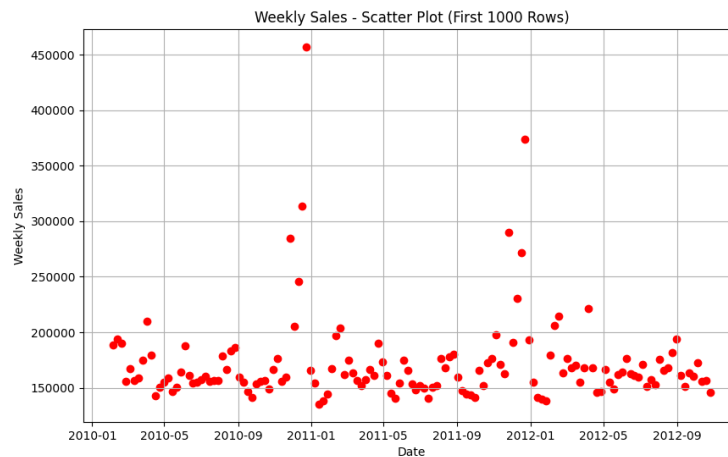


Figure 5 — Scatter Plot

These visualizations provided essential insights into data variability, seasonality patterns, and overall sales dynamics, informing model expectations.

The ARIMA model was implemented using the **ARIMA(1,1,1)** configuration, a commonly used structure for modeling linear time-series patterns. The model was trained on the 114 training observations, and a 29-week forecast was generated to match the test period. Predicted values were aligned with actual test dates for evaluation.

LSTM is a deep learning model designed to capture nonlinear and long-range dependencies. Preprocessing for LSTM: Data scaled to (0,1) using MinMaxScaler , sliding windows created (sequence length= 5), Reshaped data into 3D structure: (samples, timesteps, features). The LSTM architecture included(One LSTM layer with **32 units** , A Dense output layer with **1 neuron**, **Adam** optimizer and **MSE** loss function ,Training for **30 epochs** with a batch size of **32**) After training, predictions were inverse transformed to return to the original sales scale.

Both models were evaluated using **MAE** (Mean Absolute Error), **MSE** (Mean Squared Error), **RMSE** (Root Mean Squared Error). These metrics were chosen because they are widely used in time-series forecasting and provide interpretable error magnitudes.

Results and Discussion:

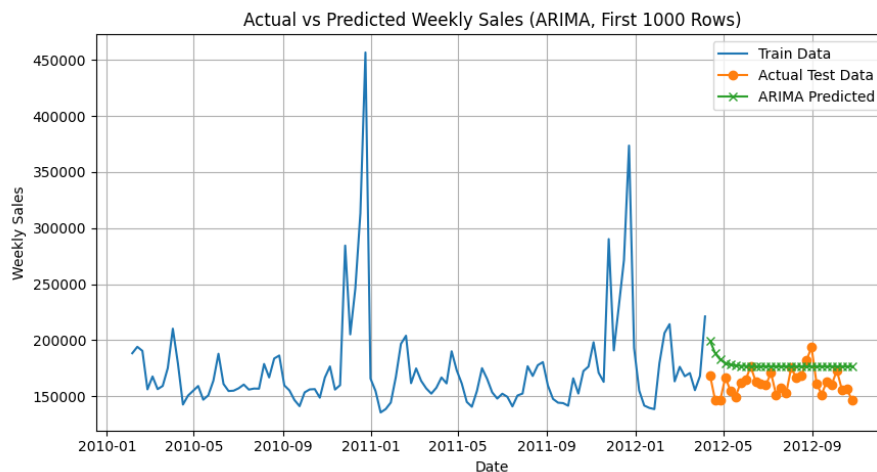
The models were evaluated on a 29-week test period using MAE, MSE, and RMSE. Table 6 summarizes the quantitative results

Metric	ARIMA	LSTM
MAE	17,531.14	12,107.87
MSE	407,182,066.80	199,682,689.36
RMSE	20,178.75	14,130.91

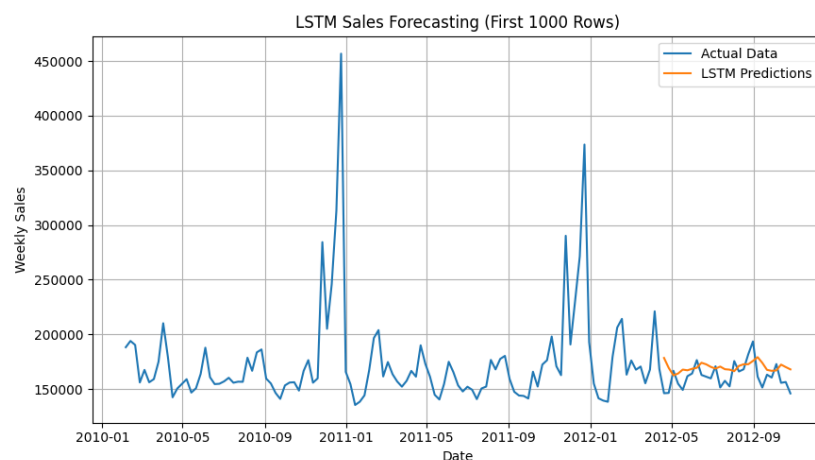
Table 3. Forecasting Performance of ARIMA vs. LSTM

The LSTM model achieves lower error values across all metrics, indicating significantly higher predictive accuracy.

Visual comparison between the actual and predicted weekly sales shows:



ARIMA closely follows the general direction of the trend but smooths fluctuations and struggles with sharp changes.



LSTM produces predictions that more closely track the movement of actual sales, especially during weeks with higher variability. These plots help verify that LSTM captures underlying nonlinear patterns more effectively.

Quantitative Interpretation

- LSTM outperforms ARIMA by a **31% reduction in MAE** and approximately **30% reduction in RMSE**.
- The significantly lower MSE confirms that LSTM handles sharp deviations and rapid shifts more effectively.

Qualitative Interpretation

- ARIMA provides a stable, interpretable baseline but is limited by linear assumptions.
- LSTM adapts better to real-world sales behavior influenced by nonlinear factors such as promotions, seasonality, and store-level variability.

Both models behave as expected for their methodological class: ARIMA performs well on linear patterns, while LSTM excels on complex sequences.

The primary research objective was to compare classical and deep-learning forecasting methods for aggregated retail sales. The results clearly demonstrate that:

- **LSTM is more effective** for capturing the complexities of Walmart's weekly sales.
- **ARIMA still serves as a valuable baseline model**, providing interpretability and quick computation for managerial scenarios.

This confirms that deep learning approaches offer measurable benefits in forecasting accuracy when sufficient data is available.

Accurate forecasting holds significant strategic value for Walmart and similar retailers. The findings suggest that LSTM-based forecasting can:

- Improve **inventory management** by reducing stockouts and overstocking
- Support **operational planning**, including staffing and scheduling
- Enhance **financial forecasting** for budgeting and performance evaluation
- Optimize **supply chain coordination**, especially during high-demand periods

These improvements translate into reduced costs, better customer satisfaction, and more informed decision-making.

Comparison with Prior Studies

- Prior retail forecasting studies often show that ARIMA is suitable for short-term, linear trends, while neural networks outperform traditional models on larger datasets.
- The results of this study are consistent with the literature: LSTM provides superior accuracy in complex retail sales data.

Limitations

- The subset used (1,000 aggregated observations) is smaller than the full dataset; using the full dataset may further improve LSTM results.
- External factors such as holidays, promotions, and economic indicators were not included, potentially limiting predictive richness.
- LSTM requires more computation and hyperparameter tuning compared to ARIMA.

And last thing Summary of Key Findings is LSTM achieved the lowest prediction error across all metrics and is the more suitable model for forecasting Walmart's aggregated weekly sales; ARIMA remains useful as a transparent baseline but lacks the flexibility needed for complex retail behaviors; The findings support the adoption of deep learning techniques in data-driven retail decision-making.

Conclusion:

This research is intended to assess and rank different statistical and modern deep learning-based approaches regarding their predictive performance for Walmart's weekly sales forecasting. In pursuit of this goal, a structured methodology has been pursued that encompasses data aggregation, exploratory analysis, model development, and performance evaluation. Both ARIMA and LSTM were trained using an aggregated subset of 1,000 weekly observations with a chronological 80/20 split for testing. The main findings show that while ARIMA gave a reliable and interpretable baseline model, the LSTM network had much better predictive performance, as evidenced from the lower MAE, MSE, and RMSE values. This underlines that LSTM is more capable of capturing nonlinear patterns and fluctuations present in retail sales data. This project contributes to the understanding of how different forecasting approaches behave when applied to real-world retail datasets and underlines the added value of deep learning models to enhance the accuracy of sales forecasting. On the whole, the research questions were addressed: the implementation of the forecasting models was done; their evaluation and comparison were carried out; and useful conclusions were obtained on their strengths and limitations. The insights from the study can help retailers in making better decisions on inventory planning, demand management, and operational forecasting.

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